Boolean-Oracle Unitary Synthesis: Contracted Feedback, Sparse Fields, and Sharper Resource Bounds

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Abstract

We analyze the pinned October 5, 2026 manuscript *Polynomial-Time Unitary Synthesis from a Boolean Oracle* through its actual unitary feedback block, finite Fourier fields, row-sparse encoders, and single-call recursion. We prove an exact feedback residual identity and a finite-period Poisson-kernel tail estimate, retaining noncommutative factor order and phase aliasing. A square-root amplification schedule reduces the encoder's repetition count without changing its programmable-slot model. A logarithmic truncation depth, together with the full inherited Solovay–Kitaev recurrence, gives sharper upper bounds for this particular construction: at fixed accuracy, its slot count is $O((n+1)^4 \log^{5/2}(n+1))$ and its workspace is $O((n+1)^4 \log^2(n+1))$. These are source-relative resource improvements, not a new abstract synthesis theorem or an efficient way to construct the Boolean truth table. We distinguish proved finite-dimensional identities, inherited existence and compilation arguments, and bounded numerical checks. The complete source text was read; this is not a formal certification of its global theorem. Contribution assessment: C3/5, a proved quantitative refinement of the source's feedback-field and programmable-slot construction, with its global oracle-existence dependencies explicitly retained.

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1 Problem, quantifiers, and evidence boundaries

Let $n \geq 1$, $N = n + 1$, and let $\mathcal{H}_d$ denote the span of the first d computational basis states in an n-qubit register, $1 \leq d \leq 2^n$. All unqualified norms are spectral operator norms. Channel errors use the *full* diamond norm, without a factor $1/2$. The source's headline assertion has the quantifier order

$$\exists \text{ uniform polynomial-size oracle-circuit family } (A_n) \quad \forall U \in U(2^n) \quad \exists f : \{0,1\}^{a(n)} \to \{0,1\}. \tag{1}$$

Its circuit, with the oracle $O_f|x, b\rangle = |x, b \oplus f(x)\rangle$, approximates the channel of U to a constant diamond error at most $1/2$. The generator takes 1^n, not a classical description of U. Neither efficiently constructing f nor efficiently evaluating it without oracle access is asserted. The table may have exponentially many entries in a polynomial address width. Counting each query as one operation is essential. Replacing that access by explicit classical memory changes the resource model.

The induction treats coherent families U_y, with a label register y. It requires the norm estimate on an initialized workspace

$$\|C\iota - \iota U_{\mathrm{ctl}}\| \leq \delta, \qquad U_{\mathrm{ctl}} = \sum_y |y\rangle\langle y| \otimes U_y, \tag{2}$$

uniformly on valid inputs and label superpositions. Here ι appends zero workspace; spectator registers may be tensored in. A unitary extension outside $\mathcal{H}_d$ is allowed, but cannot change the action on valid inputs. Scalar phases depending on y are physical relative phases, not an equivalence relation in (2).

What is inherited and what is proved here. The source supplies a contracting-block existence construction, exact slot skeletons, and a recursive synthesis scheme. Sections 3– 7 prove the local identities, sharper encoder bound, and resource consequences used here. The final resource theorem is an implication within that inherited construction, not an independent replacement for every combinatorial and uniformity argument in the source. In particular, no finite matrix test can verify (1) for every n and U.

Literal reading. All ten text files in the pinned `build` directory were read in full: `main.tex` (61 lines), `references.tex` (120), `figures/recursion.tex` (35), and sections 01–07 with respectively 168, 133, 217, 248, 186, 163, and 268 lines: 1,599 lines total. The selected private proof ranges are recorded in Section 8. Hashes, endpoint comparisons, test outcomes, and export receipts belong to the accompanying M567 evidence files. Reading breadth is not a claim of mechanized verification or a guarantee of byte immutability throughout reading.

2 Programmable slots and the inherited contraction

A programmable slot applies an arbitrary 2×2 unitary W_y to a target qubit, controlled by specified other register positions. The skeleton lists these positions, not an explicit exponentially large multiplexor. The target is not among the controls. Tables can depend on the target unitary and on all inherited labels.

The source's exact primitives are important to later costs. A diagonal operation uses one slot, with a zero-initialized qubit carrying a label-only phase when necessary. A state on u qubits is prepared in $u+1$ slots by conditional prefix rotations followed by phases; zero-probability prefixes admit arbitrary unitary completions. A permutation is implemented in $5u$ slots and u temporary qubits: compute its value into a clean register, erase the original with the inverse permutation, and swap registers. The positive Fourier transform on 2^b points uses $O(b)$ slots, grouping all the controlled phases for a given Hadamard stage into one table. This is a slot bound, not an $O(b)$ bound in a fixed elementary gate set before oracle compilation.

For each $d > 1$ the source sets

$$M = 4096N^2, \quad q = 2^{\lfloor \log_2 d \rfloor}, \quad s = \max\{1, \lfloor q/M \rfloor\}, \quad k = d - s, \quad r = 3/4. \tag{3}$$

It constructs slot unitaries P_L, P_R so that

$$V = P_L U P_R = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad A : \mathbb{C}^k \to \mathbb{C}^k, \quad D : \mathbb{C}^s \to \mathbb{C}^s, \quad \|D\| \le r. \tag{4}$$

The two permutations are not interchangeable; implementing U from V requires P_R^* first and P_L^* last.

Actual probabilistic argument. For $s = 1$, an entry in the last column has modulus at most $d^{-1/2} < 3/4$; a row permutation makes it the lower-right entry. For $s > 1$, let W be the last s columns of the normalized q-dimensional Walsh matrix, and let M_0 be the relevant contraction submatrix. Independent column signs can be fixed so that $C' = M_0 E_2 W$ has entries bounded by $4\sqrt{N/q}$ and $\|C'\| \le 1$. The union bound in the source is $4qs \exp(-4N) < 1$. Writing rows of W, C' as h_i, c_i, the proposed corner is $\sum_i \epsilon_i h_i^* c_i$. Hermitian dilation gives

$$L_i = \begin{pmatrix} 0 & h_i^* c_i \\ c_i^* h_i & 0 \end{pmatrix}, \qquad \left\| \sum_i L_i^2 \right\| \le 16sN/q.$$

The two diagonal variance blocks are bounded separately, using $W^*W = I_s$, $\|h_i\|^2 = s/q$, and $\|C'\| \le 1$; replacing them by a scalar entrywise estimate would lose the argument. The source proves, for Hermitian $h \times h$ matrices and integer $a \ge 1$,

$$\mathbb{E} \left\| \sum_i \epsilon_i L_i \right\| \le 2h^{1/(2a)} \sqrt{(2a-1) \left\| \sum_i L_i^2 \right\|}.$$

Its Gaussian integration-by-parts proof bounds the $2a$-th trace moment by $h(2a-1)!!v^a$, then uses the mean absolute Gaussian to compare signs. With $h = 2s$, $a = N$, the resulting expectation is at most $16N\sqrt{s/q} \le 1/4$. Thus signs exist. The Walsh transforms, signs, and alignment permutations cost $O(n)$ slots. This is not an algorithm for discovering the signs from an efficiently given arbitrary U.

Since $s \geq d/(4M)$, successive dimensions decrease by a factor at most $1 - 1/(4M)$ until they reach one. There are fewer than

$$J = 8MN = 32768N^3 \tag{5}$$

nontrivial levels: $2^n \exp(-J/(4M)) < 1$. The one-dimensional terminal operation is a label-dependent phase slot. This dimension bound is independent of the field cutoff chosen below.

3 Exact feedback and physical residuals

Fix the unitary block (4). For any diagonal unitary Z on $\mathbb{C}^s$, define

$$X = (I - DZ)^{-1}C, \qquad F = A + BZX, \qquad E = \begin{pmatrix} I \\ ZX \end{pmatrix}, \qquad O = \begin{pmatrix} F \\ X \end{pmatrix}. \tag{6}$$

The inverse exists in operator norm because $\|DZ\| \leq r < 1$. This is a specific internal feedback problem of a unitary colligation, not a prescription to invert an arbitrary matrix with a quantum oracle.

Proposition 3.1 (Feedback conservation and dual field)**.** *F is unitary. With $R = OF^* = [I; Y]$ one has*

$$Y = (I - Z^*D^*)^{-1}Z^*B^*, \qquad FE^* = R^*V. \tag{7}$$

Proof. The internal equation gives $VE = O$. Consequently, $I + X^*X = E^*E = O^*O = F^*F + X^*X$, so $F^*F = I$. The matrix F is square, hence also $FF^* = I$. Thus $R = OF^*$ has upper block I. From $V^*R = EF^*$ the lower block equation is $B^* + D^*Y = ZY$, yielding the stated formula. Finally $VE = RF$ implies $E^*V^* = F^*R^*$; multiplication by F on the left and V on the right gives the exact intertwining identity. No commutation of D with Z has been used. □

Proposition 3.2 (Residual and conservation defect)**.** *For any $\widehat{X} : \mathbb{C}^k \to \mathbb{C}^s$, put $\mathcal{R}_0 = C - (I - DZ)\widehat{X}$ and $\widehat{F} = A + BZ\widehat{X}$. Then*

$$X - \widehat{X} = (I - DZ)^{-1}\mathcal{R}_0, \tag{8}$$

$$F - \widehat{F} = BZ(I - DZ)^{-1}\mathcal{R}_0, \tag{9}$$

$$\widehat{F}^*\widehat{F} - I = -\widehat{X}^*\mathcal{R}_0 - \mathcal{R}_0^*\widehat{X} - \mathcal{R}_0^*\mathcal{R}_0. \tag{10}$$

In particular $\left\|F - \widehat{F}\right\| \leq \|\mathcal{R}_0\| / (1 - r)$. For $X_L = \sum_{l=0}^{L-1}(DZ)^l C$, its residual is exactly $(DZ)^L C$.

Proof. Subtract the defining internal equations to get (8); multiply by BZ to get (9). On a vector u, unitarity of V gives

$$V \begin{pmatrix} u \\ Z\widehat{X}u \end{pmatrix} = \begin{pmatrix} \widehat{F}u \\ \widehat{X}u + \mathcal{R}_0 u \end{pmatrix}.$$

Equality of the squared norms for every u proves (10). The Neumann series bounds the resolvent by $(1 - r)^{-1}$, and $\|B\| \leq 1$. The last assertion follows by telescoping the finite geometric sum. □

The sign of the quadratic residual term is fixed by the original energy identity. An approximate inverse or a small coefficient residual without the factors BZ does not certify the implemented channel. Likewise, replacing $\widehat{F}$ by its polar factor may repair a matrix's unitarity while destroying the row sparsity and slot access required by the algorithm.

4 Finite phase fields and their exact defects

Let $L \geq 1$ be an integer and set

$$b = nL + \lceil \log_2(L+1) \rceil, \quad P = 2^b, \quad G = (\mathbb{Z}/P\mathbb{Z})^{L+1}, \quad K = b(L+1). \tag{11}$$

For internal indices $1 \leq i \leq s$, put $v_i = (1, i, \ldots, i^L)$ and $Z(z) = \mathrm{diag}(\chi_{v_i}(z))$, where $\chi_v(z) = \exp(2\pi \mathrm{i}\langle v, z\rangle/P)$. Writing $z = (z_0, z')$ gives $Z = \omega Z'$ with $\omega^P = 1$. We have $P > L$ and, more strongly, $P > L2^{nL}$.

Define finite fields

$$E_L(z) = \begin{pmatrix} I \\ \sum_{l=0}^{L-1} Z(DZ)^l C \end{pmatrix}, \qquad R_L(z) = \begin{pmatrix} I \\ \sum_{l=0}^{L-1} (Z^*D^*)^l Z^* B^* \end{pmatrix} \tag{12}$$

and analysis maps from $\mathbb{C}^{k+s}$ to $\mathbb{C}^G \otimes \mathbb{C}^k$ by

$$\mathcal{E}_L x = |G|^{-1/2} \sum_z |z\rangle E_L(z)^* x, \qquad \mathcal{R}_L x = |G|^{-1/2} \sum_z |z\rangle R_L(z)^* x.$$

Let $\mathcal{F} = \sum_z |z\rangle\langle z| \otimes F(z)$.

Proposition 4.1 (Original physical Gram matrices)**.** *For $W_E = DZ'$ and $W_R = Z'^* D^*$,*

$$\mathcal{E}_L^* \mathcal{E}_L = \mathrm{diag}\left(I_k, I_s - \mathbb{E}_{z'} Z' W_E^L (W_E^*)^L Z'^*\right), \tag{13}$$

$$\mathcal{R}_L^* \mathcal{R}_L = \mathrm{diag}\left(I_k, I_s - \mathbb{E}_{z'} W_R^L (W_R^*)^L\right). \tag{14}$$

In particular $(1 - r^{2L})I \leq \mathcal{E}_L^* \mathcal{E}_L, \mathcal{R}_L^* \mathcal{R}_L \leq I$.

Proof. Average first over z_0. The lower terms of E_L have distinct positive degrees $1, \ldots, L$, and the lower terms of R_L distinct negative degrees $-1, \ldots, -L$. They are pairwise orthogonal modulo P; they also have zero average against the degree-zero upper block. In the forward case, unitarity gives $CC^* = I - DD^* = I - W_E W_E^*$. Thus the lower covariance is

$$Z' \sum_{l=0}^{L-1} W_E^l (I - W_E W_E^*)(W_E^*)^l Z'^* = Z'\big(I - W_E^L (W_E^*)^L\big) Z'^*.$$

In the backward case set $C_R = Z'^* B^*$. Then $C_R C_R^* = I - W_R W_R^*$, using $B^*B + D^*D = I$. The same telescope, without the external Z' factors, proves (14). Positivity and $\|W_E\|, \|W_R\| \leq r$ give the inequalities. □

The exact defect quantities

$$d_E = \left\|\mathbb{E}_{z'} Z' W_E^L (W_E^*)^L Z'^*\right\|, \qquad d_R = \left\|\mathbb{E}_{z'} W_R^L (W_R^*)^L\right\| \tag{15}$$

can replace r^{2L} in encoder error estimates if bounded uniformly over all coherent external labels. This changes the analysis of the original fields, not their coefficients or Fourier row support. Computing those averages by enumerating all labels is not generally efficient.

4.1 Exact finite-period tail covariance

The infinite fields E, R from Proposition 3.1 exist at every point of this same finite group. They are not assumed to have Gram matrix I: degrees separated by P alias. This distinction is essential.

Theorem 4.2 (Poisson-kernel tail bound)**.** *For a contraction W with $\|W\| \le r < 1$, suppose $CC^* = I - WW^*$. For $L \ge 1$ and integer $P > L$, set $T(\omega) = \sum_{l=L}^{\infty} \omega^{l+1} W^l C$ on the P-th roots. With $A = W^P$, $H = (I - A)^{-1}$, one has exactly*

$$\mathbb{E}_\omega T(\omega) T(\omega)^* = W^L H (I - AA^*) H^* (W^*)^L, \tag{16}$$

$$H(I - AA^*)H^* = H + H^* - I. \tag{17}$$

This positive covariance has norm at most $r^{2L}(1 + r^P)/(1 - r^P)$. Consequently

$$\|\mathcal{F}\mathcal{E}_L - \mathcal{R}_L V\| \le 2r^L \sqrt{\frac{1 + r^P}{1 - r^P}}. \tag{18}$$

Proof. Absolute operator-norm convergence permits grouping the series by $l = L + v + aP$, $0 \le v < P$, $a \ge 0$. Its v-th coefficient is $W^{L+v} HC$, since H commutes with every power of W. The P distinct residue frequencies are orthogonal. Their covariance sums to

$$W^L H \left(\sum_{v=0}^{P-1} W^v C C^* (W^*)^v \right) H^* (W^*)^L = W^L H (I - W^P (W^*)^P) H^* (W^*)^L.$$

Only functions of W have been commuted with W; neither C nor W^* has been commuted across W. To prove (17), multiply both sides on the left by $I - A$ and on the right by $I - A^*$: the right side becomes $(I - A^*) + (I - A) - (I - A)(I - A^*) = I - AA^*$. Also

$$H + H^* - I = I + \sum_{a \ge 1} (A^a + (A^*)^a), \qquad \|H + H^* - I\| \le 1 + 2 \sum_{a \ge 1} r^{aP} = \frac{1 + r^P}{1 - r^P}.$$

Apply this covariance result at each z' with W_E, C, conjugating by Z' for $E - E_L$. For $R - R_L$ use W_R, C_R; changing all phase powers to negative ones preserves residue orthogonality and the same covariance. Averaging over z' preserves the positive upper bounds, so both analysis map tails have norm at most $r^L \sqrt{(1 + r^P)/(1 - r^P)}$. Finally $\mathcal{F}\mathcal{E} = \mathcal{R}V$ pointwise, and $\mathcal{F}, V$ are unitaries. The triangle inequality proves (18). □

For comparison the source bounds each tail pointwise by $r^L/(1-r)$. The finite-period improvement is genuinely tied to its averaged fields. Discarding the factor involving r^P would require an additional limiting argument and is not justified by replacing discrete phases with continuous ones. The exact covariance also applies to nonnormal W.

4.2 Why these fields, rather than arbitrary inverse columns

After a Fourier transform of the G register, each row of $\mathcal{E}_L$ or $\mathcal{R}_L$ has at most L nonzero input entries. Here is the actual support argument. A monomial of degree $m \le L$ has frequency $\sum_{a=1}^{m} v_{i_a}$. Each coordinate of this sum is an integer at most $L 2^{nL} < P$, so equality modulo P implies ordinary integer equality. The first coordinate determines m. The next m power sums determine the elementary symmetric polynomials by Newton's identities, and hence the multiset of internal indices, including multiplicities. In the adjoint forward field, the input internal index occurs as the rightmost conjugated

diagonal phase; in the backward adjoint field it occurs as the rightmost unconjugated phase. Either way the input index must belong to this frequency multiset. There are at most $m \le L$ possibilities. The upper identity block contributes only at frequency zero, where each row has a single input entry. Internal terms have nonzero degree modulo P, so they do not meet that row. The different signs of the two Fourier supports do not change their row-sparsity bound.

This is a support theorem, not a polynomial-time algorithm to recover all row entries. The slot tables can contain their values and completion choices. A generic Schur complement inverse need not have any comparable Fourier support structure; transferring just a Neumann estimate would miss the implementability mechanism.

5 A shorter row-sparse encoder

We now give a source-relative improvement to the repetition schedule. The mechanism is ordinary amplitude amplification [2]; the result below supplies the precise clean-input error and cost needed by these fields. It does not claim a new amplitude-amplification theorem.

Let $T : \mathbb{C}^d \to \mathbb{C}^{2^w}$ have at most $L \ge 1$ nonzeros per row and

$$(1-\epsilon)I \le T^*T \le I, \qquad 0 \le \epsilon < 1. \tag{19}$$

The d valid inputs fit in n qubits. Put $h = \lceil \log_2(L+1) \rceil$. There are fixed embeddings ι for input with zero workspace and j for output rows with zero input-index register and both success flags one. They refer to the same layout in the forward and backward encoders.

Lemma 5.1 (Attenuated sparse block)**.** *For any $t_0 > 0$ with $t_0\sqrt{L} \le 1$, a slot unitary B_0 can be chosen with*

$$j^* B_0 \iota = t_0 T. \tag{20}$$

It uses $O(n+w+h)$ slots and $w+h+2$ fresh qubits in the source layout. All choices can be coherent in external labels with a common skeleton.

Proof. For each valid column v, conditionally prepare the row register and first flag so that its flag-one amplitudes are T_{uv}. This is possible because $\sum_u |T_{uv}|^2 \le 1$; place the remaining probability in the flag-zero subspace. For each row u, choose a list I_u of exactly L distinct extended input indices containing all its nonzero valid columns. There are enough indices since $2^{n+h} > L$; padding entries carry zero matrix coefficients. Conditionally apply the inverse of a preparation of $L^{-1/2}\sum_{v\in I_u}|v\rangle$ to the index register. Its zero-index amplitude is $1/\sqrt{L}$ on each listed input. A rotation of the second flag with amplitude $t_0\sqrt{L}$ gives (20). All other amplitudes remain in orthogonal failure subspaces, and all steps have unitary completions. Conditional state preparation costs the number of target qubits plus one in this slot model. The extra h index qubits, the row register, and two flags give the stated workspace. None of these tables is asserted to be efficiently computable from T. □

Theorem 5.2 (Square-root repetition schedule)**.** *In Lemma 5.1, choose*

$$m = \lceil \sqrt{L} \rceil, \qquad \theta = \frac{\pi}{4m+2}, \qquad t_0 = \sin\theta. \tag{21}$$

There is a unitary encoder U_T satisfying

$$\|U_T\iota - jT\| \le 2\big(1-\sqrt{1-\epsilon}\big) \le 2\epsilon, \tag{22}$$

with $O(\sqrt{L}(n+w+h))$ slots and the same fresh workspace as above.

Proof. First, $t_0\sqrt{L} \le \theta\sqrt{L} \le \pi/4 < 1$, so the attenuated block is available. Let $P_0 = \iota\iota^*$ and $Q_0 = jj^*$. The encoder is

$$U_T = \big[(2B_0P_0B_0^* - I)(I - 2Q_0)\big]^m B_0.$$

For singular vectors $Tv_a = \sigma_a e_a$, define $\beta_a = \arcsin(t_0\sigma_a)$ and $\widetilde{e}_a = je_a$. Then $B_0\iota v_a = \sin\beta_a\,\widetilde{e}_a + \cos\beta_a\,e'_a$, where the e'_a are orthonormal and orthogonal to the entire good subspace. This follows by subtracting the projected components in (20) from the orthonormal vectors $B_0\iota v_a$. Each singular plane is invariant under the two reflections, whose product increases this angle by $2\beta_a$. After m repetitions the angle is $(2m+1)\beta_a$.

Since $(2m+1)\theta = \pi/2$, put $\Delta_a = (2m+1)(\theta - \beta_a) \ge 0$. The derivative of $\arcsin(\sigma\sin\theta)$ is at most $\tan\theta$ on $0 \le \sigma \le 1$. Also $m \ge 1$ and $\theta \le \pi/6$, so

$$\Delta_a \le (2m+1)\tan\theta(1-\sigma_a) \le \sqrt{3}(1-\sigma_a).$$

For the last inequality, $\tan x/x$ increases on $(0,\pi/2)$: the derivative's numerator $x\sec^2 x - \tan x$ starts at zero and has derivative $2x\sec^2 x\tan x \ge 0$. Its value at $x = \pi/6$ gives $(\pi/2)(\tan x/x) \le \sqrt{3}$ on this interval. The squared error against the desired vector $\sigma_a\widetilde{e}_a$ is

$$1 + \sigma_a^2 - 2\sigma_a\cos\Delta_a = (1-\sigma_a)^2 + 2\sigma_a(1-\cos\Delta_a) \le (1-\sigma_a)^2 + \sigma_a\Delta_a^2 \le 4(1-\sigma_a)^2.$$

Distinct singular planes are orthogonal. Taking the maximum over $\sqrt{1-\epsilon} \le \sigma_a \le 1$ proves (22). Reflections about the prescribed zero-index and flag subspaces are diagonal slot operations; each repetition calls B_0, B_0^* once. Thus the total cost is $O(m(n+w+h))$. Label control makes this a direct sum of the same argument, preserving all scalar phases and the operator norm bound. □

The source uses $m = L$, $\theta = \pi/(4L+2)$, and a 5ϵ bound. The shorter schedule is possible because the preparation needs only $t_0\sqrt{L} \le 1$, not $t_0 L \le 1$. For the fields, $w = K + n$ and $\epsilon = r^{2L}$, or separately d_E, d_R from (15). Fourier transforms add $O(K)$ slots; unitary changes of the row basis do not change the Gram bounds. This improvement is an upper bound within the given slot construction, not a lower-bound or optimality claim.

6 Single-call recursion and clean error transport

Suppose a smaller controlled unitary $F(z)$ has an implementation with clean error δ' on the shared good embedding j. Its action may be arbitrary outside that initialized subspace. Let U_E, U_R be encoders for $\mathcal{E}_L, \mathcal{R}_L$ on a common workspace, with errors e_E, e_R. The larger step is $U_R^* C_F U_E$, with fresh inner-recursion workspace.

Proposition 6.1 (One recursive call)**.** *The clean error relative to V is at most*

$$\delta' + e_E + e_R + 2r^L\sqrt{\frac{1+r^P}{1-r^P}}. \tag{23}$$

With Theorem 5.2, one may take $e_E = e_R = 2(1-\sqrt{1-r^{2L}})$.

Proof. The first encoder differs from $j\mathcal{E}_L$ by at most e_E. Its error is propagated through unitaries, so its norm does not grow. On $j\mathcal{E}_L$, the smaller implementation differs from $j\mathcal{F}\mathcal{E}_L$ by at most δ' because $\|\mathcal{E}_L\| \le 1$ and its port register is valid. Theorem 4.2 then compares this to $j\mathcal{R}_L V$. Finally $\|U_R^* j\mathcal{R}_L - \iota\| \le e_R$, by multiplying the encoder's own error by U_R^*. Add the three comparisons. The argument never assumes that the actual intermediate state has perfectly reset workspace. □

Only one call to C_F occurs. Amplification repeats the nonrecursive sparse preparations, not C_F. Otherwise the depth-J recurrence could be exponential. The conditioning unitaries P_R^*, P_L^* and their exact slot implementations add no approximation error at this stage.

For $r = 3/4$, $P > L \geq 1$, the elementary estimate $\sqrt{(1+r^P)/(1-r^P)} \leq 1/(1-r) = 4$ yields

$$e_E + e_R + 2r^L \sqrt{\frac{1+r^P}{1-r^P}} \leq 4r^{2L} + 8r^L \leq 12r^L. \tag{24}$$

The sharper Poisson estimate is useful for constants or variable depths; the conservative last bound is sufficient for the asymptotic ledger.

Lemma 6.2 (Hybrid bound with reusable scratch)**.** *Let the ideal slot sequence preserve its required clean subspace at each boundary. Suppose its a-th compiled unitary has error at most η_a on that ideal boundary subspace. On initially clean inputs the entire sequence has error at most $\sum_a \eta_a$.*

Proof. In the telescoping difference, write each term as an actual unitary suffix, one actual-minus-ideal slot, and an ideal prefix. The prefix reaches the subspace on which the local bound is stated; the suffix has norm one on all states, including leaked scratch states. Apply the triangle inequality. This argument also holds with untouched spectators. □

Lemma 6.3 (From clean isometries to channels)**.** *If isometries W_1, W_2 have $\|W_1 - W_2\| \leq a$, their channels, after discarding any workspace, have diamond distance at most $2a$.*

Proof. For a pure input with any reference register, subtract its two output outer products by inserting one mixed outer product. The trace norm of $|u\rangle\langle v|$ is $\|u\|\,\|v\|$, so the result is at most $2a$. Convexity handles mixed inputs, and partial trace is contractive on Hermitian differences. Stabilization does not change an operator norm. □

7 Logarithmic cutoffs and compiled resource bounds

The source takes $L = 10000J$ and bounds $(3/4)^L$ by a weak reciprocal estimate. No encoder or Fourier-support hypothesis requires this linear dependence on J. All require only integer $L \geq 1$ and the definitions in (11).

Proposition 7.1 (Logarithmic depth)**.** *For a desired recursion error $0 < \delta < 1$, the choice*

$$L = \left\lceil \frac{\log(12J/\delta)}{\log(4/3)} \right\rceil \tag{25}$$

with the square-root encoders gives total clean recursion error at most δ. Retaining the original encoders instead, any integer $L > \log(1800J)/\log(4/3)$ gives error less than $1/100$.

Proof. Sum (24) over fewer than J levels and use $(3/4)^L \leq \delta/(12J)$. With the original encoders the source step bound is $2r^L/(1-r) + 10r^{2L} \leq 18r^L$; the second assertion follows identically. In either case the terminal phase is exact before compilation. □

For rational δ the generator can avoid transcendental rounding: find the smallest positive integer L with $12J3^L \leq \delta 4^L$ using integer multiplication after clearing the denominator. This gives (25) up to the harmless equality convention. Its cost is polynomial in the binary sizes of the rational precision parameter and the resulting L. At the fixed accuracy of the source all these are functions of n alone.

Let t denote the number of initial label qubits. From the source layout, each of the fewer than J levels adds $O(K+n+h)$ workspace and uses $O(\sqrt{L}(K+n+h)+K+n)$ slots after Theorem 5.2. Here $b=O(NL)$, $K=O(NL^2)$ and $h=O(\log(L+1))$. Thus

$$Q = t + O(N^4L^2), \qquad S = O(N^4L^{5/2}). \tag{26}$$

No product of the dimensions of the successive label registers enters these counts: a slot may be controlled by all their positions, with an arbitrary table. Describing each control list does cost $O(Q)$ bits or operations up to index encoding factors. It remains polynomial and must not be confused with writing the table itself.

7.1 Keeping the full inherited commutator recurrence

The source's phase-sensitive compilation first proves density of determinant-one words in $H, T, T^\dagger$, then uses a fixed finite ε_0-net with maximum word length ℓ_0. Its commutator step already gives

$$\varepsilon_{j+1} \le C\varepsilon_j^{3/2}, \qquad \ell_j \le 5^j\ell_0. \tag{27}$$

The source weakens this to halving the error, and hence a ξ^{-3} word bound. Retaining the recurrence gives the standard Solovay–Kitaev exponent [3], not a new approximation theorem.

For clarity, the recurrence has a phase-sensitive justification. For an $SU(2)$ residual $SW_0^* = \exp(\mathrm{i}a\cdot\sigma)$ with $|a| = O(\varepsilon)$, choose skew-Hermitian P, Q of norm $O(\sqrt{\varepsilon})$ and $[P,Q] = \mathrm{i}a\cdot\sigma$. Perpendicular Pauli vectors with cross product $-a/2$ do this. The commutator of e^P, e^Q agrees with the residual to $O(\varepsilon^{3/2})$, by expanding the exponential series through degree two and bounding its norm-convergent cubic tail. Approximating e^P, e^Q to ε by the net changes the commutator by at most

$$2\varepsilon\left\|e^Q - I\right\| + 2\varepsilon\left\|\widetilde{e^P} - I\right\| = O(\varepsilon^{3/2}).$$

The identity terms cancel in each conjugation difference. Multiplying by W_0 uses five old words. Inverses use the same alphabet. Every word remains determinant one. The initial density/net assertion is inherited; it uses the irrational rotation with trace $1+\sqrt{2}/2$ and a second nonparallel rotation axis, not an approximation modulo scalar phase.

Lemma 7.2 (Word length from the recurrence)**.** *There is a constant K_0 such that any $0<\xi<1$ can be attained with word length at most*

$$K_0(1+\log(1/\xi))^\kappa, \qquad \kappa = \frac{\log 5}{\log(3/2)} < 4. \tag{28}$$

A larger constant times $(1+\lceil\log_2(1/\xi)\rceil)^4$ is an integer-computable uniform length bound.

Proof. Enlarge C to at least one, and choose the fixed net so $x_0 = C^2\varepsilon_0 < 1$. Put $x_j = C^2\varepsilon_j$. Then $x_{j+1} \le x_j^{3/2}$, hence $x_j \le x_0^{(3/2)^j}$. For sufficiently small ξ, it suffices that $(3/2)^j \ge \log(1/(C^2\xi))/\log(1/x_0)$. The least such nonnegative integer j has $5^j = O((1+\log(1/\xi))^\kappa)$. All remaining ξ are absorbed by increasing K_0. Since $\kappa<4$, the second assertion follows after adjusting the constant for log base and integer rounding. □

7.2 One Boolean oracle, actual phases, and exact erasure

Write a slot entry $W_y = e^{i\gamma_y} S_y$, $S_y \in SU(2)$. Approximate both S_y and $\mathrm{diag}(e^{i\gamma_y}, e^{-i\gamma_y})$ to $\eta/2$ by the words above; apply the second to a phase qubit initially zero. Their ideal product is W_y with that qubit still zero, and their error is at most η. Pad all words with identities to the same length ℓ. One Boolean function contains the two symbol bits for each tuple

$$(\text{slot},\ y,\ \text{first/phase word},\ \text{position},\ \text{bit}).$$

The y field is padded to Q bits. The address width is $a = O(Q + \log(S+1) + \log(\ell+1))$. Copying the selected controls, setting constant address bits, and undoing these steps requires $O(a)$ elementary operations per symbol position. Two oracle calls read the two symbol bits, and two erase them. There are two words, so at most $8S\ell$ queries suffice. Slot controls never change during a target operation, making the oracle uncomputation exact even on coherent labels.

The finite alphabet can be selected with a constant number of allowed gates, not arbitrary controlled continuous rotations. In particular,

$$4abc = a + b + c - (a \oplus b) - (a \oplus c) - (b \oplus c) + (a \oplus b \oplus c)$$

implements the doubly controlled Z phase with $T, T^\dagger$ on computed parities. Hadamard conjugation yields Toffoli. Controlled T uses a computed AND and its exact uncomputation. For controlled H, set $D_0 = (T^2H)T(T^2H)^*$; then $D_0 Z D_0^* = H$, including its phase. Apply D_0^*, controlled Z, then D_0. These constructions justify the constant per-symbol selection overhead and $O(a)$ address overhead.

The word selections and their scalar phases are stored in the oracle; the circuit generator never selects them. The fixed constant in Lemma 7.2 may be hardcoded as in the source. This proves existence of a polynomial generator, not a useful explicit numerical gate-count constant. Lemma 6.2 proves total compilation error $S\eta$ even when the phase qubit is reused and its actual state has small leakage.

Theorem 7.3 (Source-relative improved ledger)**.** *Retain the source's contracting-block and exact-slot construction. For $n \geq 1$ and $0 < \varepsilon_{\mathrm{ch}} \leq 1$, let $\delta = \varepsilon_{\mathrm{ch}}/4$, choose L by (25), and compile with $\eta = 1/h_0$ where $h_0 = \lceil 4(S+1)/\varepsilon_{\mathrm{ch}} \rceil$. Put $B = 1 + \log((S+1)/\varepsilon_{\mathrm{ch}})$. Then the resulting channel has full diamond error at most $\varepsilon_{\mathrm{ch}}$, and upper bounds are*

$$Q = t + O(N^4 L^2), \qquad S = O(N^4 L^{5/2}), \tag{29}$$

$$\textit{oracle calls} = O(SB^4),$$

$$\textit{elementary gates} = O\big(SB^4[Q + \log(S+1) + \log(B+1)]\big), \tag{30}$$

$$\textit{total qubits, address bits} = O\big(Q + \log(S+1) + \log(B+1)\big). \tag{31}$$

For fixed accuracy and $t = 0$, these become respectively

$$\begin{aligned} Q &= O(N^4 \log^2 N), \qquad & S &= O(N^4 \log^{5/2} N), \\ \textit{calls} &= O(N^4 \log^{13/2} N), \qquad & \textit{gates} &= O(N^8 \log^{17/2} N). \end{aligned} \tag{32}$$

Proof. The recursion contributes at most $\varepsilon_{\mathrm{ch}}/4$ and the compiled slots at most $S/h_0 < \varepsilon_{\mathrm{ch}}/4$ in clean operator norm. Lemma 6.3 gives the channel conclusion. Equation (26) gives (29), while Lemma 7.2 gives $\ell = O(B^4)$. The lookup counts just established give (30)–(31). At fixed accuracy, $L = O(\log N)$ and $\log(S+1) = O(\log N)$, giving (32). Integer loops over the skeleton, word positions, and address positions write the circuit in polynomial time; binary gate indices add only encoding factors to a bit-level output-time bound. For the original fixed-accuracy theorem the generator takes only 1^n. Variable accuracy requires a rational precision parameter or an integer reciprocal as an additional input, with its representation length accounted for. □

Fixed-accuracy slot construction	Workspace beyond labels	Slots
Source's $L = 10000J$, $m = L$	$O(N^{10})$	$O(N^{13})$
Logarithmic L, original $m = L$	$O(N^4 \log^2 N)$	$O(N^4 \log^3 N)$
Logarithmic L, $m = \lceil \sqrt{L} \rceil$	$O(N^4 \log^2 N)$	$O(N^4 \log^{5/2} N)$

The table compares upper bounds derived from the same construction. It is not a measured runtime gain, a lower-bound separation, or a claim that the oracle's full description has polynomial length. The fourth-power log used for word lengths is intentionally a simple upper bound; retaining κ instead would slightly improve the displayed logarithmic powers.

8 Private proof transfers and their limits

The applicability routing is L4, indicating strong mathematical overlap; it is not a C4 contribution assessment. The routes actually followed were the core finite-section/error discussion (map lines 104–118), foundations physical-metric and analytic-inverse discussions (83–111), and application structured-inverse discussion (177–188). The following original proofs, not just their titles, were read locally. Their confidential contents were not used in web searches.

Structured inverse and original residual. The original `TR-core-pat-5.tex` was read at lines 175–240 and 310–436. These ranges include the nilpotent/Neumann arguments and the Schur residual and Woodbury discussions. For $A = D_0(I - M_0)$, $\|M_0\| \le q < 1$, the truncated inverse has a norm-convergent geometric remainder, while the physical residual is $D_0 M_0^{K+1} D_0^{-1} b$, not a coordinate-free power of M_0 alone. The exact Schur lift similarly includes the interior residual term $A_{BI} A_{II}^{-1} r_I$. Proposition 3.2 specializes this discipline to the actual feedback equation: the port error includes BZ, and its conservation defect includes the quadratic residual term. It does not import an arbitrary low-rank inverse into the synthesis model. The Chebyshev material in the same read scope has normal/self-adjoint spectral hypotheses and is not applied to the generally nonnormal DZ.

Physical metric. The original Volume 21 `ch01d-physical-metrics.tex`, lines 1–160, includes the proofs for an injective frame J_0, its metric $G_0 = J_0^* J_0$, and the orthogonal projector $J_0 G_0^{-1} J_0^*$. For a self-adjoint physical operator, the coefficient operator $G_0^{-1} J_0^* A J_0$ is metric-Hermitian, not automatically Euclidean Hermitian. The full residual contains the omitted orthogonal component. Here the physical Gram matrices are exactly (13)–(14). Using their defects in (15) preserves the original fields and their sparse Fourier supports. Entrywise smallness or a new artificial metric would not supply the encoder hypothesis (19) in the circuit's Hilbert norm.

Analytic inverse: an actual norm hypothesis. The original `TR-rev-1-old-tr147.tex`, lines 1794–1900, includes the complete analytic-inverse proposition and proof at line 1836 onward. Its Banach-space Neumann argument requires the actual kernel operator on the specified analytic-jet space to have norm below one. Formal triangular invertibility alone is insufficient. The finite-dimensional counterpart used here is $\|DZ\| < 1$ in the physical operator norm. For example,

$$T_0 = \begin{pmatrix} 1/2 & 4 \\ 0 & 1/2 \end{pmatrix}$$

has spectral radius $1/2$ but $\|T_0\| > 1$. Spectral radius controls eventual convergence in finite dimensions, but does not give the source's uniform geometric bound with ratio $3/4$ or the Gram

contractions used in its recursion. The private proof supplies no automatic circuit access to an inverse and no theorem about oracle-table constructibility.

These transfers explain why the useful reformulation is a physical feedback residual and a finite-phase covariance calculation. The new resource ledger additionally depends on the actual slot encoder, which does not follow from any of these general private inverse results.

9 Public context and source status

The analyzed source is the October 5, 2026 public-title manuscript pinned to the OpenAI math repository commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a` [1]. The accompanying provenance record checks selected local bytes against the available October 8 captured Git trees. Such a comparison is a dated source-identity check, not a fresh live-branch proof review. Neither unchanged bytes nor the repository's hosting imply a theorem has been independently certified. No source revision is inferred merely from a title search.

The public primary literature was refreshed on October 8, 2026, using public titles and public bibliographic identifiers only. Amplitude amplification [2] and the Solovay–Kitaev construction [3] are established mechanisms; the changes here concern their parameters and accounting within this source. Lombardi, Ma, and Wright's one-query lower bound [4] rules out the corresponding one-query universal synthesis proposal, not every multi-query scheme. It should not be cited as a direct contradiction of a polynomial-query assertion.

The current primary arXiv record for *All Unitaries Have Constant Depth Quantum Circuits* is version 2, October 5, 2026, by Nehoran, Slote, and Yuen [5]. Its abstract states polynomial-depth exact implementation with $2^{O(n)}$ ancillas, and constant depth when unbounded fan-out is allowed. Those width and gate-model allowances matter: a small depth bound with exponentially many ancillas is not by itself the polynomial-width Boolean-oracle theorem studied here. This report uses that record for current context, not as a replacement proof of the pinned source or as support for a historical priority claim about the present resource refinements.

10 Benefits, costs, and further work

10.1 Benefits of the source-specific reformulation

The exact residual formula tests the implemented port operator, including the order of its feedback factors. The Gram identities keep the actual Hilbert-space error separate from coefficientwise approximations. The Poisson-kernel tail calculation uses the finite common-phase register already present in the construction, with no extra circuit and no loss of row sparsity. It identifies precisely the small aliasing correction that a continuous-phase argument would omit.

The cutoff improvement is especially inexpensive: it changes an integer parameter already free in the source, while leaving the contraction and single-call architecture unchanged. The square-root encoder schedule also uses the same preparations and flags, changing only the attenuation and number of reflections. Keeping the full commutator recurrence avoids an unnecessary polynomial precision penalty. Together these yield explicit resource bounds under exactly the oracle-table convention at issue, rather than a generic claim that an inverse is efficiently approximable.

10.2 Costs and unresolved dependencies

The constants in the contraction construction are large: $J = 32768N^3$ already precedes any field or compilation overhead. The finite-net word constant is an existence constant, not an implemented

compiler constant. The width of the phase register remains $O(NL^2)$ per level. Uniform table lookup is coherent and unit-cost only because it is assigned to the oracle; storing or constructing every table explicitly can dominate all circuit costs. Nothing here provides an efficient classical synthesis algorithm for arbitrary input matrices.

The conditioning signs, sparse-row completion tables, exact preparation tables, and recursively controlled unitaries are inherited existence choices. Their compatibility is analyzed along the stated architecture, but this report is not a machine-checked global proof of all dimensions. Testing finitely many blocks cannot establish that arbitrary labels and all recursive completion choices satisfy the global construction. Nor does a smaller proven upper bound show that actual running times are smaller than another algorithm's, or that the bound is optimal.

There are also genuine analytic boundaries. Letting $r \uparrow 1$ worsens the truncation depth and resolvent sensitivity. If only $\rho(DZ) < 1$ is known, nonnormal transient growth can invalidate every uniform r^L estimate used here. Replacing $\|D\| \le r$ by a spectral statement therefore needs quantitative power bounds and a replacement for the positive telescoping Gram argument, not a new notation for the same proof. Approximate unitarity of V perturbs the telescopes by accumulated energy defects; no such perturbation is silently absorbed in the present theorem.

10.3 Concrete extensions and obstacles

Nonuniform cutoff allocation. If level j has a certified $r_j < 1$ and uniformly bounded exact defects $d_{E,j}, d_{R,j}$, Proposition 6.1 permits the explicit budget

$$\sum_j \left[2r_j^{L_j} \sqrt{\frac{1+r_j^{P_j}}{1-r_j^{P_j}}} + 2(1-\sqrt{1-d_{E,j}}) + 2(1-\sqrt{1-d_{R,j}}) \right] \le \delta.$$

Each P_j must still satisfy the multiset no-wrap bound at its chosen L_j. Minimizing slot cost under this sufficient budget could lower constants. The obstacle is obtaining bounds uniform over exponentially many coherent labels without enumerating them. This formula is a proved sufficient condition, not a claim that the optimizing allocation or the underlying exact errors are efficiently computable.

Approximate colligations. A practical extension would start from a numerical block matrix with certified $\|V^*V - I\|$ and a strict contraction margin for D. One must rederive both row and column energy defects, the feedback nonunitarity, and the corrected field Gram bounds. Positive lower and upper Gram bounds may then allow a modified encoder. Simply taking a polar factor of the whole field is inadequate because it can densify Fourier rows and invalidate the slot-count proof.

More economical frequency codes. The moment vector $(1, i, \dots, i^L)$ ensures uniqueness by power sums but uses $L+1$ coordinates with large modulus. A shorter additive multiset code could reduce K and hence both width and compiled gate count. The required property is injectivity for every multiset of size at most L, together with a separable phase action and the common-phase orthogonality used in the Gram proof. A generic hashing argument with collisions would not preserve the exact row-sparsity theorem.

Explicit table generation and constants. For structured families of U, derandomizing the sign choices and computing preparation tables may turn oracle existence into an actual uniform compilation algorithm. That requires additional structure of U and bit-complexity bounds for every

recursive label; no such structure is assumed in (1). An explicit finite-net construction would also make the constants in the retained Solovay–Kitaev bound quantitative. Neither extension follows from the local tests below.

11 Contribution and verification scope

The realized assessment is C3, separate from applicability L4. The substantial source-relative content consists of the exact finite-period covariance proof (Theorem 4.2), the shorter encoder with its clean error bound (Theorem 5.2), and the combined logarithmic-cutoff and polylogarithmic-compilation ledger (Theorem 7.3). Propositions 3.1 and 3.2 expose original-operator identities supporting this analysis; the basic feedback conservation law itself is inherited. The cutoff and full-recurrence observations were also identified in the parallel source review. No historical novelty is assigned to Neumann series, amplitude amplification, or Solovay–Kitaev refinement.

The M567 sidecars bind the final scientific source to reproducible local checks of block conservation, residuals, nonnormal finite-phase covariance, encoder singular-plane errors, integer cutoff inequalities, phase-sensitive gate identities, and resource accounting. Their actual pass/fail outcomes and numerical tolerances are recorded there, not inferred from compiling this report. Floating-point checks illustrate and challenge the derived formulas; the proofs above, rather than finite samples, justify their stated quantifiers. They do not test a synthesized oracle for arbitrary n. The full-text reading ledger is broader than the independent algebraic test scope. The export and visual records concern the final PDF's physical rendering, not theorem validity.

References

[1] OpenAI math repository, *Polynomial-Time Unitary Synthesis from a Boolean Oracle*, October 5, 2026, pinned manuscript and ten build text files. Pinned public manuscript.

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[3] C. M. Dawson and M. A. Nielsen, *The Solovay–Kitaev algorithm*, arXiv:quant-ph/0505030v2, 2005.

[4] A. Lombardi, F. Ma, and J. Wright, *A one-query lower bound for unitary synthesis and breaking quantum cryptography*, arXiv:2310.08870, 2023.

[5] B. Nehoran, J. Slote, and H. Yuen, *All Unitaries Have Constant Depth Quantum Circuits*, arXiv:2609.40351v2, October 5, 2026.